Chapter 6

Semiconductors

INTRODUCTION

In the previous chapter we examined the band theory of solids and saw how it led to an understanding of the differing electrical characteristics among solids. In this chapter we shall examine some of the properties of semiconductors. As we have seen, at absolute zero a perfect crystal of normally semiconducting material would be an insulator. The characteristic of semiconductivity is caused by thermal vibrations which disrupt the perfect order of the crystal and generate mobile charge carriers. The semiconductor materials are just that, i.e., their conductivities are intermediate between good conductors (10^{6} ohm^{-1} cm^{-1}) and good insulators (10^{-15} ohm^{-1} cm^{-1}). The conductivity of semiconductors ranges from about 10^{-8} to 10^{2} ohm^{-1} cm^{-1}.

INTRINSIC SEMICONDUCTORS

To begin our study of semiconductors we must first distinguish the behavior of a very pure material from a less pure one. It is a characteristic of semiconductors that their electrical properties are greatly influenced by the presence of impurities even in amounts as small as one part per million and less. Those semiconductors that are very pure and whose behavior is not dominated by impurity atoms are called *intrinsic semiconductors*.

At temperatures very near absolute zero an intrinsic semiconductor can be characterized by a completely empty energy band separated by a small energy gap from a completely filled band as seen in Fig. 6-1. The filled band is made up of the electronic states of the outer or valence electrons of the atoms and is called the *valence band*. The forbidden region or band gap is ordinarily about 1 eV.

As the temperature of the crystal is raised, some electrons will be excited across the forbidden region into the empty upper band. There will be a continuous process of electron transfer both into the upper band from below and out of the upper band and into the

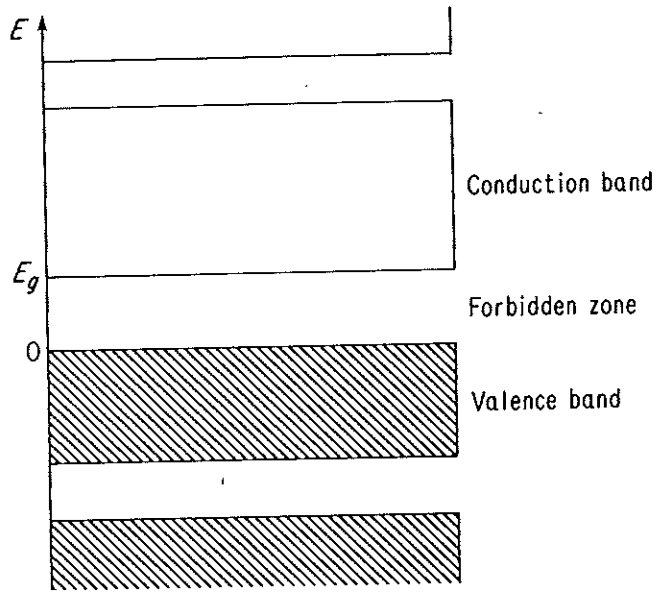


Fig. 6-1. Energy-band picture near valence band in a semiconductor.

valence band. At a given temperature the number in the upper band will stabilize when the rates of transfer into and out of the band become equal. Since the electrons in this nearly empty band have available states nearby, they can be made to carry a current. For this reason the upper band is called the *conduction band.*

The transfer of electrons out of the valence band is equivalent to the introduction of holes into that band. These holes can also contribute to an electric current. Since all of the electrons excited into the conduction band create holes in the valence band, their numbers will be exactly equal. If we let n be the electron density in the conduction band and p be the hole density in the valence band, we have

$$n_i = p_i \tag{6-1}$$

where the subscripts indicate intrinsic materials.

In determining the conductivity we must include the effects of carriers of both kinds. Thus we write

$$\sigma = |e|\,(n\mu_n + p\mu_p) \tag{6-2}$$

where μ_n and μ_p represent the electron and hole mobilities. Both mobilities and the electronic charge are all taken as positive, since the two contributions to the conductivity are additive. Note, however, that the actual drift velocities will be in opposite directions.

Since the carrier concentration and hence the conductivity of a semiconductor are orders of magnitude smaller than those of a metal,

we are safe in assuming that the difference in energy between the bottom of the conduction band and the chemical potential, or Fermi level, μ is much greater than kT for temperatures of interest. If we measure energies from the top of the valence band, then the lowest energy of the conduction band is E_g, the magnitude of the forbidden gap. Under the condition that

$$E_g - \mu \gg kT \tag{6-3}$$

the Fermi-Dirac distribution function becomes

$$f(E_g) \simeq e^{-(E_g-\mu)/kT} = e^{-\Delta E/kT} \tag{6-4}$$

If the mobilities are slowly varying functions of temperature, then the temperature dependence of the conductivity will be due to changes in the carrier concentration. Near the bottom of the conduction band the electrons behave as nearly free electrons and we may describe their density of states with the formula developed in Chapter 4. The only requirement is that the energy should be measured from the bottom of the band. Since we have already decided to use the top of the valence band as an energy reference, we must replace the energy in Eq. 4-87 with an effective energy $E' = E - E_g$. Thus we get

$$G_n(E)\,dE = \frac{1}{2\pi^2}\left(\frac{2m_n^*}{\hbar^2}\right)^{3/2}(E - E_g)^{1/2}\,dE \tag{6-5}$$

for the number of states between E and $E + dE$ per unit volume. For energies within the energy gap, $G(E)\,dE$ is zero.

The number of occupied states per unit volume is the electron concentration given by

$$n = \int_{E_g}^{\infty} G_n(E)\,f_n(E)\,dE \tag{6-6}$$

where the upper limit of the integral has been extended to infinity for sake of convenience. The subscripts denote that $G(E)$ and $f(E)$ are appropriate for electrons. Upon inserting Eq. 6-5 into Eq. 6-6, we get

$$n = \frac{1}{2\pi^2}\left(\frac{2m_n^*}{\hbar^2}\right)^{3/2} e^{\mu/kT}\int_{E_g}^{\infty}(E - E_g)^{1/2}e^{-E/kT}\,dE \tag{6-7}$$

Making the change of variable from E to $E' = E - E_g$ the integral above becomes

$$e^{-E_g/kT}\int_0^{\infty} E'^{1/2}e^{-E'/kT}\,dE' = \frac{(\pi kT)^{3/2}}{2\pi}\,e^{-E_g/kT} \tag{6-8}$$

Upon combining Eqs. 6-7 and 6-8 we find the electron density to be

$$n = 2\left(\frac{2\pi m_n^* kT}{h^2}\right)^{3/2} e^{(\mu - E_g)/kT} \tag{6-9}$$

In a similar fashion we may calculate the density of positive carriers. For these we measure kinetic energies as positive downward from the top of the valence band. Thus the appropriate density of states function is

$$G_p(E)\,dE = \frac{1}{2\pi^2}\left(\frac{2m_p^*}{\hbar^2}\right)^{3/2} (-E)^{1/2}\,dE \tag{6-10}$$

The hole concentration is given by

$$p = \int_{-\infty}^{0} G_p(E) f_p(E)\,dE \tag{6-11}$$

where the Fermi function for holes is given by

$$f_p(E) = 1 - f_n(E) \simeq e^{(E-\mu)/kT} \tag{6-12}$$

Upon combining Eqs. 6-10 through 6-12, the hole concentration is shown to be

$$p = 2\left(\frac{2\pi m_p^* kT}{h^2}\right)^{3/2} e^{-\mu/kT} \tag{6-13}$$

If we examine the product np of the electron and hole concentrations, we find

$$np = 4\left(\frac{2\pi kT}{h^2}\right)^3 (m_n^* m_p^*)^{3/2} e^{-E_g/kT} \tag{6-14}$$

This is a most important and useful result. The product np is independent of the position of the Fermi level μ. This means that for a particular crystal, in which the energy gap and effective masses are fixed, the product np depends only on the temperature. Furthermore, the product np is independent of the density of atoms in the crystal. As a result, if alterations are made to the crystal which leave the bands unchanged but which increase the electron density then the changes will necessarily cause a decrease in the hole concentration. An increase in the electron density in the conduction band would be accompanied by an increased rate of transfer back to the valence band, thereby reducing the number of holes. As we shall see in the next section, this result leads to some especially important applications in controlling the number and types of charge carriers in a semiconductor crystal.

For an intrinsic material the number of electrons will equal the number of holes. By Eq. 6-14 we see that

$$n_i = p_i = (n_i p_i)^{1/2} = (np)^{1/2} \tag{6-15}$$

We now have a way of determining the value of the Fermi level μ. Upon equating n and p of Eqs. 6-9 and 6-13, we find that

$$e^{2\mu/kT} = (m_p^*/m_n^*)^{3/2}\, e^{E_g/kT} \tag{6-16}$$

Equivalently,

$$\mu = \tfrac{1}{2}E_g + 3/4\; kT \ln (m_p^*/m_n^*) \tag{6-17}$$

For $kT \ll E_g$ the Fermi level will be approximately in the center of the forbidden zone. If the effective masses are equal, then $\mu = \frac{1}{2}E_g$ exactly. More importantly, we see that for $E_g \gg kT$, our original assumption of $E_g - \mu \gg kT$ was indeed valid. This result is appropriate for the real semiconductor crystals where the band gaps encountered vary from 0.23 ev to 3 ev as seen in Table 6-1. For room temperature of approximately 300° K, the value of kT is 1/40 eV, which assures the inequalities required above.

Table 6-1
Energy Gap between Valence and Conduction Bands in Some Common Semiconductors at Room Temperature (eV)*

Si	1.14	PbTe	0.30
Ge	0.67	CdS	2.42
InSb	0.23	ZnS	3.6
GaAs	1.4		
SiC	3		
PbS	0.34		

*C. Kittel, *Introduction to Solid State Physics*, 3d ed. (New York: Wiley, 1966), p. 302.

As a final variant we may write Eqs. 6-9 and 6-13 in the following form:

$$n = N_c\, e^{(\mu - E_g)/kT} = N_c e^{-\mu/kT} \tag{6-18}$$

and

$$p = N_v\, e^{-\mu/kT} \tag{6-19}$$

where N_c and N_v are the effective densities of energy levels in the conduction and valence bands respectively. N_c and N_v are seen to be slowly varying functions of the temperature. For an intrinsic

material,

$$n_i = p_i = (N_v N_c)^{1/2} e^{-E_g/2kT} \tag{6-20}$$

For silicon the value of n_i at room temperature is 1.4×10^{10} cm^{-3}. The value of n_i for germanium is about 2,000 times larger due to the smaller size of the band gap. The room temperature n_i for Ge is 2.4×10^{13} cm^{-3}.

In the valence bands of silicon and germanium there are two parabolic bands that are nearly degenerate at $k = 0$. Since their curvatures are unequal these bands contain holes of unequal masses. These light and heavy holes are present simultaneously in the crystal. In the conduction band the constant energy surfaces are ellipsoidal. Consequently there are two effective masses associated with the longitudinal and transverse axes of these ellipsoids. However, it is still possible to define a density of states equivalent effective mass for use in equations such as Eqs. 6-14 and 6-17. A full discussion of

Table 6-2
Some Important Properties of Germanium, Silicon, and Gallium-Arsenide at 300°K

Semiconductor	Ge	Si	GaAs
Effective density of states*			
N_c(cm^{-3})	1.04×10^{19}	2.8×10^{19}	4.7×10^{17}
N_v(cm^{-3})	6.0×10^{18}	1.04×10^{19}	7.0×10^{18}
Intrinsic carrier* concentration			
n_i(cm^{-3})	2.4×10^{13}	1.45×10^{10}	$\sim 9 \times 10^{6}$
Mobilities*			
Electron (cm^2/volt sec)	3,900	1,350	8,600
Hole (cm^2/volt sec)	1,900	480	250
Effective band mass			
m_n^*/m_e	0.22†	0.33†	0.072‡
m_p^*/m_e	0.39	0.55	0.47

*A. S. Grove, *Physics and Technology of Semiconductor Devices* (New York: Wiley, 1967) p. 102.

†R. A. Smith, *Semiconductors* (London: Cambridge U. P., 1959) pp. 347, 350.

‡H. Ehrenreich, *Phys. Rev.*, 120, 1951 (1960). m_p^* calculated from data of this reference.

this is given in the text by R. A. Smith.[1] Some representative effective masses are listed in Table 6-2.

EXTRINSIC SEMICONDUCTORS

Extrinsic or impurity semiconductors are made by doping intrinsic semiconducting material with impurity atoms of higher or lower valence. Addition of impurities with higher valence than four results in materials whose conductivities are dominated by electron carriers. Addition of atoms with fewer than four valence electrons leads to materials in which the dominant carriers are holes.

The addition of an impurity atom with five valence electrons into a crystal of silicon results in set of localized electronic energy levels. The level corresponding to the conduction band state of the host crystal actually lies slightly below the conduction band in the forbidden zone. Such impurity levels usually lie only a small fraction of an electron volt from the band edge, as seen in Table 6-3. The

Table 6-3
Experimental Ionization Energies for Donor and Acceptor Impurities in Silicon and Germanium Obtained by Thermal Methods (eV)*

Impurity	Si	Ge	
P	.044	.0120	
As	.049	.0127	Donors
Sb	.039	.0096	
B	.045	.0104	
Al	.057	.0102	Acceptors
Ga	.065	.0108	
In	.16	.0112	

*W. Kohn, *Solid State Physics*, 5, 262 (1957).

energy required to transfer an electron from an impurity level to the conduction band is so small that we expect most of the impurity sites to be empty (or ionized) at room temperature. In fact this will be true for temperatures quite below room temperature. Since the product np must be constant, we expect the material to be one in which the negative carriers will dominate, i.e., $n \gg p$. Impurity atoms which increase the number of negative carriers are called *donor impurities*. The resulting material is called an n-type semiconductor.

[1] R. A. Smith, *Semiconductors* (London: Cambridge U. P., 1959), Chaps. 4, 10.

In a similar fashion impurities with only three valence electrons lead to localized electron states only slightly above the valence band. These states would be empty at 0° K. At room temperature, however, thermal excitation of electrons into these local states removes them from the valence band and results in an excess of holes or positive carriers. These impurities are known as *acceptor impurities* since they accept or trap electrons out of the valence band. The resulting materials are called p-type semiconductors because of the dominance of the positive carriers.

Let us now examine the behavior of these impurity semiconductors in a quantitative way. For a host material of band gap E_g we denote the energies of donor and acceptor states as E_d and E_a respectively. This is illustrated in Fig. 6-2 where the impurity levels

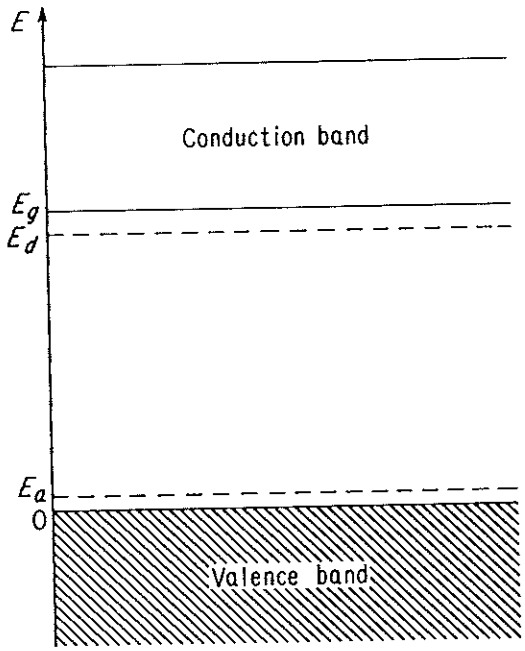


Fig. 6-2. Band gap in extrinsic semiconductor.

are indicated by the dashed lines. As before, we take the top of the valence band as the reference zero of energy.

Let N_d and N_a represent the density of donor and acceptor impurities. Then denote the density of neutral or occupied donor sites as N_d^0 and the density of ionized or empty sites as N_d^+. Similarly denote the density of neutral or empty acceptor sites as N_a^0 and the filled or ionized acceptor sites as N_a^-. Next we relate these various symbols with the help of the Fermi-Dirac distribution function.

The density of occupied donor levels is[2]

$$N_d^0 = N_d\, f(E_d) = \frac{N_d}{1 + \tfrac{1}{2}e^{(E_d-\mu)/kT}} \tag{6-21}$$

The number of ionized donors per unit volume is

$$N_d^+ = N_d\,[1 - f(E_d)] = \frac{N_d}{1 + 2e^{-(E_d-\mu)/kT}} \tag{6-22}$$

The neutral and ionized acceptor concentrations are given by

$$N_a^0 = N_a\,[1 - f(E_a)] = \frac{N_a}{1 + 2e^{-(E_a-\mu)/kT}} \tag{6-23}$$

and

$$N_a^- = N_a\, f(E_a) = \frac{N_a}{1 + \tfrac{1}{2}e^{(E_a-\mu)/kT}} \tag{6-24}$$

The condition of charge neutrality is expressed by

$$n + N_a^- = p + N_d^+ \tag{6-25}$$

where n and p are the concentrations of electrons in the conduction band and holes in the valence band as usual. If we now replace each item in Eq. 6-25 with the appropriate expression involving Fermi functions, the chemical potential is seen to be dependent on the concentrations of impurities and available states in the bands. Using Eqs. 6-18 and 6-19 we see that

$$N_c e^{(\mu-E_g)/kT} + \frac{N_a}{1 + \tfrac{1}{2}e^{(E_a-\mu)/kT}} = N_v e^{-\mu/kT} + \frac{N_d}{1 + 2e^{-(E_d-\mu)/kT}} \tag{6-26}$$

Except for the very lowest of temperatures, we expect most impurity sites to be ionized. This means that the exponential terms in N_d^+ and N_a^- must be small. In this case

$$N_c e^{(\mu-E_g)/kT} + N_a = N_v e^{-\mu/kT} + N_d \tag{6-27}$$

We can rearrange Eq. 6-27 to give

$$e^{2\mu/kT} - \frac{N_d - N_a}{N_c}\, e^{\mu/kT} e^{E_g/kT} - \frac{N_v}{N_c}\, e^{E_g/kT} = 0 \tag{6-28}$$

[2]The additional factor of ½ before the exponential term in the Fermi function is a consequence of the statistics associated with the electron's spin quantum number. Even though an electron in a donor or acceptor state may have either of two spin quantum numbers only one electron is normally allowed to occupy a particular state. This causes the Fermi-Dirac distribution functions to be modified as shown.

This is recognized as a quadratic equation for $e^{\mu/kT}$. By the usual algebraic technique we get

$$e^{\mu/kT} = \frac{N_d - N_a}{2N_c} e^{E_g/kT} + \sqrt{\left(\frac{N_d - N_a}{2N_c}\right)^2 e^{2E_g/kT} + \frac{N_v}{N_c} e^{E_g/2kT}} \tag{6-29}$$

The choice of the positive sign before the square root insures that $e^{\mu/kT}$ is positive when $N_d = N_a$. The logarithm of Eq. 6-29 may be found with the aid of the identity

$$\ln (u + \sqrt{u^2 + a^2}) = \ln a + \sinh^{-1} \frac{u}{a} \tag{6-30}$$

The identification of

$$u = \frac{N_d - N_a}{2N_c} e^{E_g/kT} \tag{6-31}$$

and

$$a = \sqrt{\frac{N_v}{N_c}} e^{E_g/2kT} \tag{6-32}$$

leads to

$$\mu = \tfrac{1}{2}E_g + kT \ln \sqrt{\frac{N_v}{N_c}} + kT \sinh^{-1} \frac{N_d - N_a}{2\sqrt{N_c N_v}} e^{E_g/2kT} \tag{6-33}$$

The first two terms in Eq. 6-33 are identical with Eq. 6-17 for the intrinsic semiconductor. In addition, we saw earlier that the density of carriers in an intrinsic material is

$$n_i = \sqrt{N_c N_v}\, e^{-E_g/2kT} \tag{6-20}$$

so that the expression for the Fermi level becomes

$$\mu = \tfrac{1}{2}E_g + kT \ln (m_p^*/m_n^*)^{3/4} + kT \sinh^{-1} \left(\frac{N_d - N_a}{2n_i}\right) \tag{6-34}$$

If the density of donor and acceptor impurities are equal, the term $kT \sinh^{-1} \left(\frac{N_d - N_a}{2n_i}\right)$ vanishes and the Fermi level is identical with that of the intrinsic material. In such a situation the donor and acceptor impurities *compensate* for each other. If the donors are more populous than the acceptors, the Fermi level will shift upward toward the conduction band. Conversely, the Fermi level will shift downward with an excess of acceptors as required by the inverse hyperbolic sine. In the *high-temperature limit* in which the number

of carriers thermally excited across the band gap exceeds the number available from impurity states, the $\sinh^{-1}$ term becomes negligible and the Fermi level becomes just that of the intrinsic material.

In other cases, however, the net impurity concentration $|N_d - N_a|$ will be considerably larger than the density of thermally generated carriers n_i from interband transitions. This situation, corresponding to moderately high temperatures and small total impurity density, is frequently observed at *room temperature.* We can estimate the function $\sinh^{-1} u$ using the relation

$$\sinh^{-1} u = \ln (u + \sqrt{u^2 + 1}) \tag{6-35}$$

For this condition of $u \gg 1$, Eq. 6-35 becomes

$$\sinh^{-1} u \simeq \pm \ln 2|u| \tag{6-36}$$

Thus for an n-type material in which $N_d - N_a \gg n_i$ we find the chemical potential to be

$$\mu = \tfrac{1}{2}E_g + kT \ln \frac{N_d}{n_i} \tag{6-37}$$

Here we have ignored the term $\frac{3}{4} kT \ln \frac{m_p^*}{m_n^*}$, since it is small compared to the other terms. Furthermore, if we take $N_c = N_v$, Eq. 6-20 becomes

$$n_i = N_c e^{-E_g/2kT} \tag{6-38}$$

This leads to a final expression for the Fermi level of

$$\mu = E_g - kT \ln \frac{N_c}{N_d} \tag{6-39}$$

for an n-type semiconductor. Similarly, the position of μ in a p-type semiconductor may be determined to be

$$\mu = kT \ln \frac{N_v}{N_a} \tag{6-40}$$

At *extremely low temperatures* we expect that the conduction band of an n-type material will contain only electrons resulting from ionized donors. The statement of charge conservation is

$$n - N_d^+ = 0 \tag{6-41}$$

Upon evaluating these terms we find

$$N_c e^{(\mu - E_g)/kT} - \frac{N_d}{1 + 2e^{-(E_d - \mu)/kT}} = 0 \tag{6-42}$$

Here again we obtain a quadratic equation for $e^{\mu/kT}$. Solving as before we get

$$\mu \simeq \tfrac{1}{2}(E_g + E_d) + \tfrac{1}{2}\, kT \ln \frac{N_d}{2N_c} - kT \sinh^{-1} \left(\sqrt{\frac{N_c}{8N_d}}\, e^{(E_d - E_g)/kT} \right) \tag{6-43}$$

which becomes simply

$$\mu = \tfrac{1}{2}(E_g + E_d) \tag{6-44}$$

at $T = 0°\,\mathrm{K}$. This is shown schematically in Fig. 6-3.

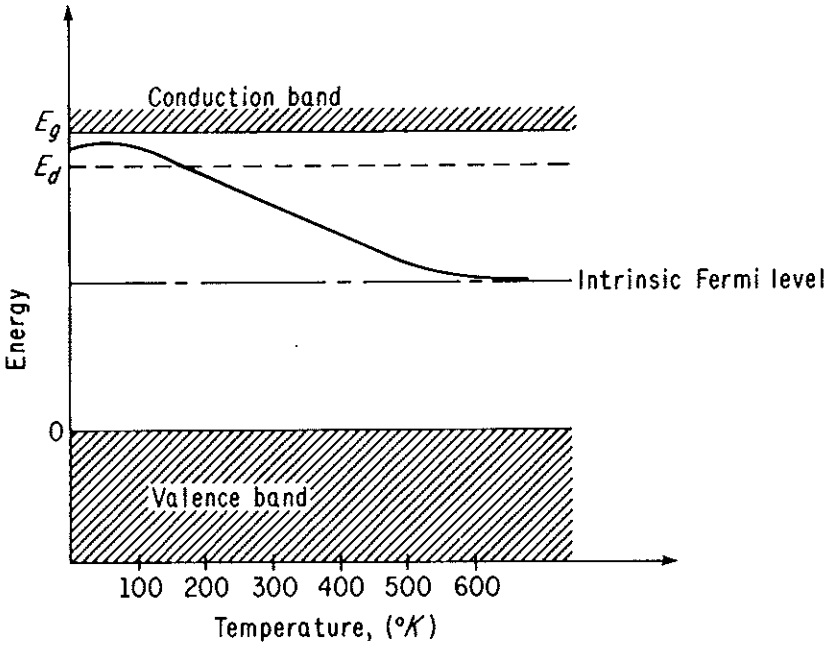


Fig. 6-3. Variation of Fermi level with temperature in an n-type semiconductor.

EXCESS CARRIER LIFETIME

In the preceding sections we have discussed the production of free carriers in semiconductors by thermal excitation of electrons into energy states which would be unoccupied at 0° K. This mechanism provides for free electrons in the conduction band and free holes in the valence band. Although the carrier density may be quite large at normal temperatures, it is still small compared to the total number of valence electrons in the crystal. Any particular electron excited into the conduction band is likely to stay there for only a short time before returning to the valence band. This process of an electron returning to the valence band with the accompanying loss of a